msvtx_chiSquared

MC vs data · overview · rank 17 / 98

rank 17 / 98 by endcap D_equal

used by: neither NN list

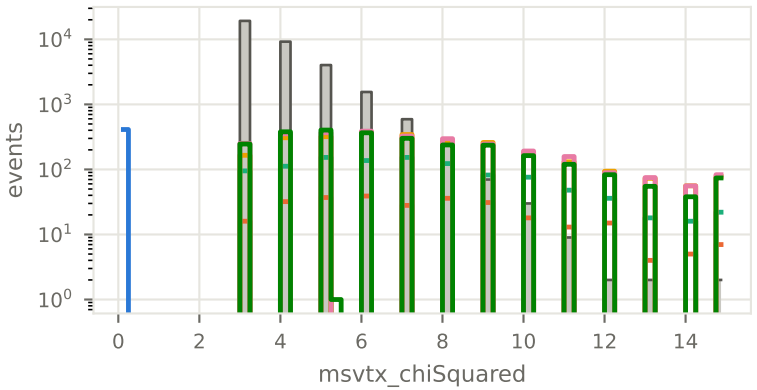
region	D_equal	noise	D_pooled	125/15	125/55	200/50	400/100	600/150	1000/275	n_bkg	rank
endcap	0.486	0.022	0.576	1.000	0.643	0.620	0.635	0.603	0.582	34,970	17/98
barrel	0.167	0.051	0.220	1.000	0.278	0.236	0.234	0.242	0.253	1,089	60/98
all	0.405	0.018	0.490	1.000	0.550	0.525	0.544	0.509	0.494	36,059	21/98

msvtx_chiSquared

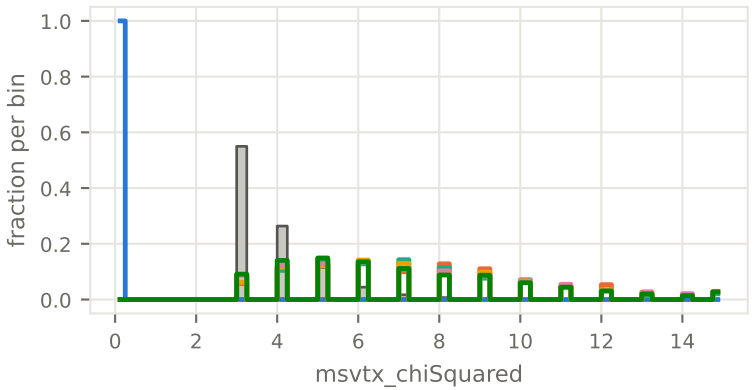
region: ENDCAP · rank 17 / 98

D_equal 0.486 (noise 0.022) D_pooled 0.576 min 0.582 (mH1000_mS275) max 1.000 (mH125_mS15) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

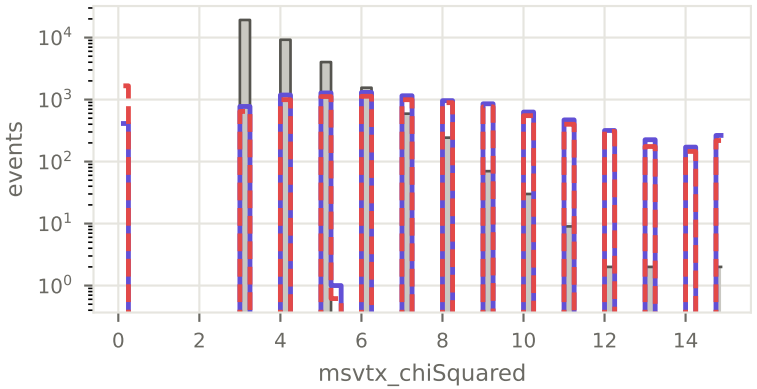


1b · each mass point vs background · normalized

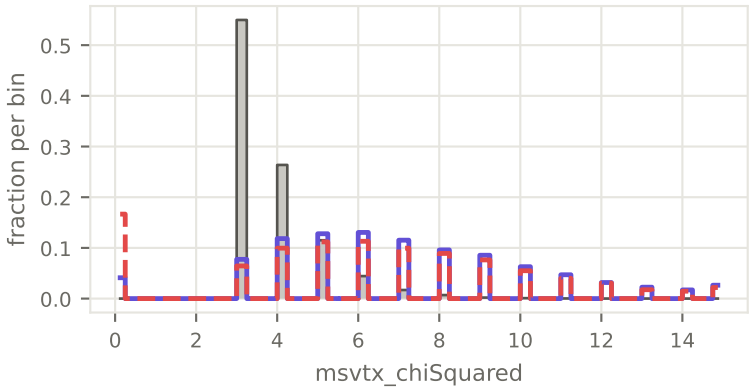


1 · each mass point
background n 34,970
mH125_mS15 D 1.000 n 411
mH125_mS55 D 0.643 n 281
mH200_mS50 D 0.620 n 1,071
mH400_mS100 D 0.635 n 2,648
mH600_mS150 D 0.603 n 2,882
mH1000_mS275 D 0.582 n 2,703

2a · combined signal vs background · raw

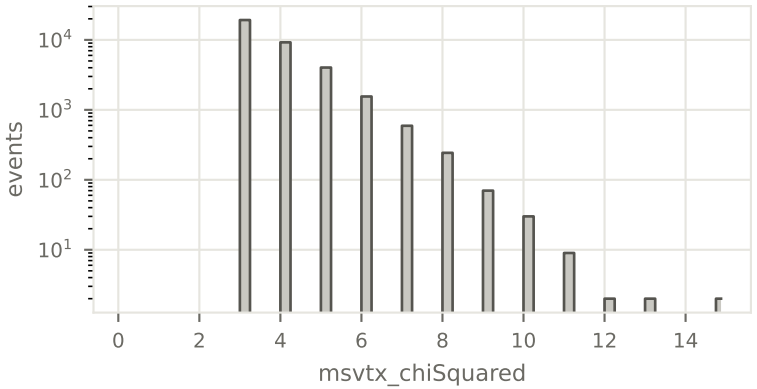


2b · combined signal vs background · normalized

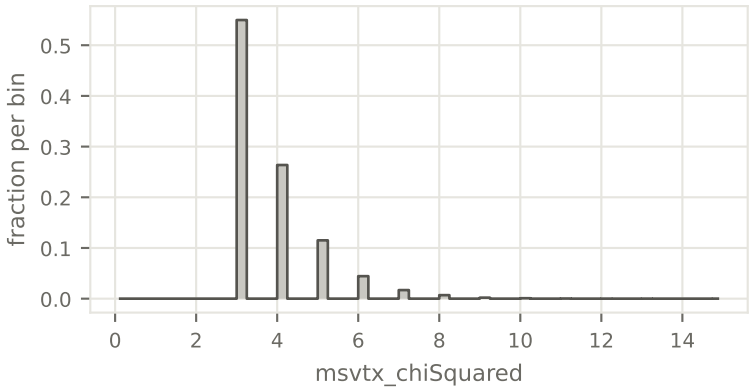


2 · combined signal
background n 34,970
combined, pooled D 0.576 n 9,996
equal weight per mass D 0.486

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

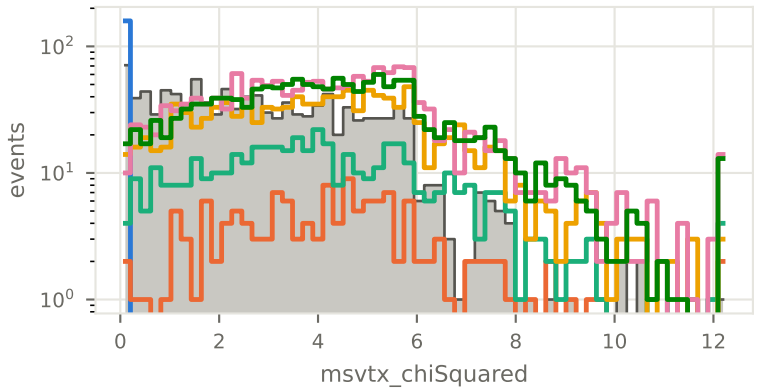
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_chiSquared

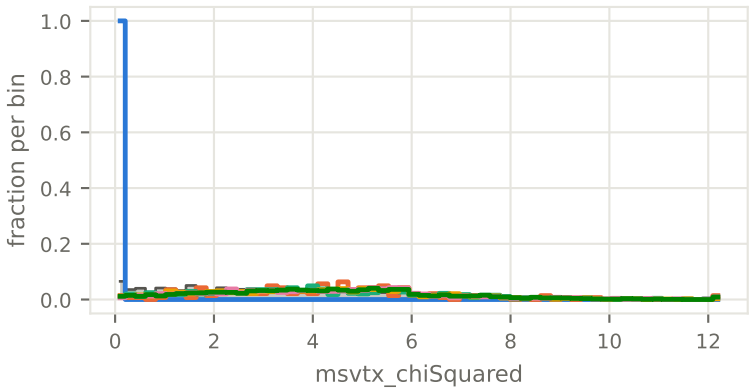
region: BARREL · rank 17 / 98

D_equal 0.167 (noise 0.051) D_pooled 0.220 min 0.234 (mH400_mS100) max 1.000 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

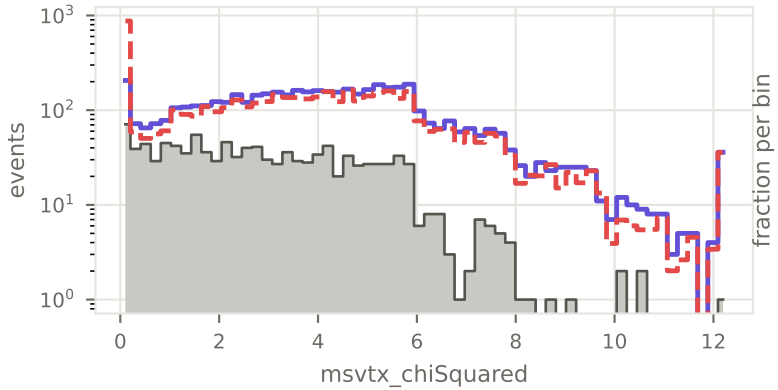


1b · each mass point vs background · normalized

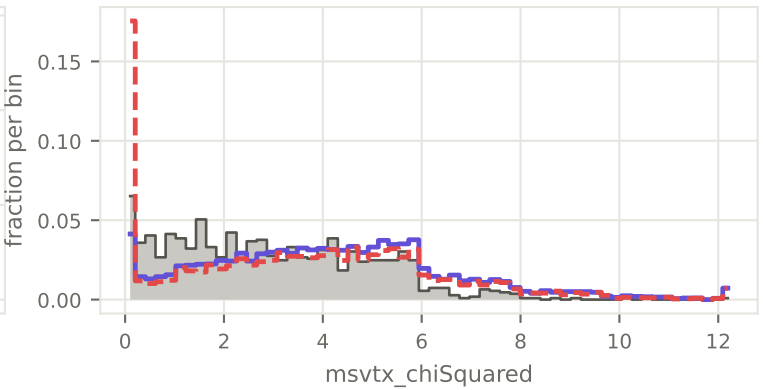


1 · each mass point
background n 1,089
mH125_mS15 D 1.000 n 159
mH125_mS55 D 0.278 n 142
mH200_mS50 D 0.236 n 449
mH400_mS100 D 0.234 n 1,144
mH600_mS150 D 0.242 n 1,608
mH1000_mS275 D 0.253 n 1,485

2a · combined signal vs background · raw

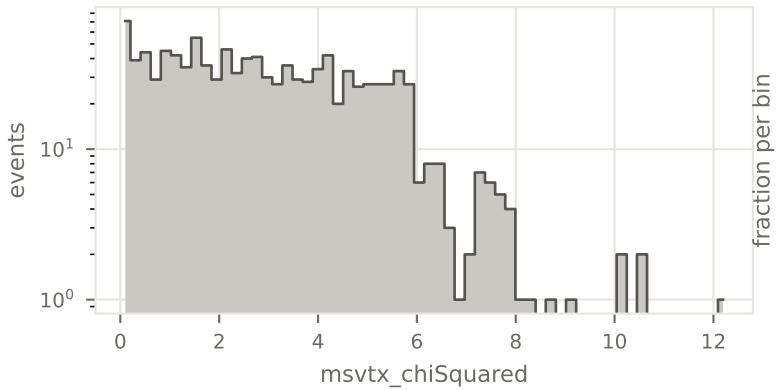


2b · combined signal vs background · normalized

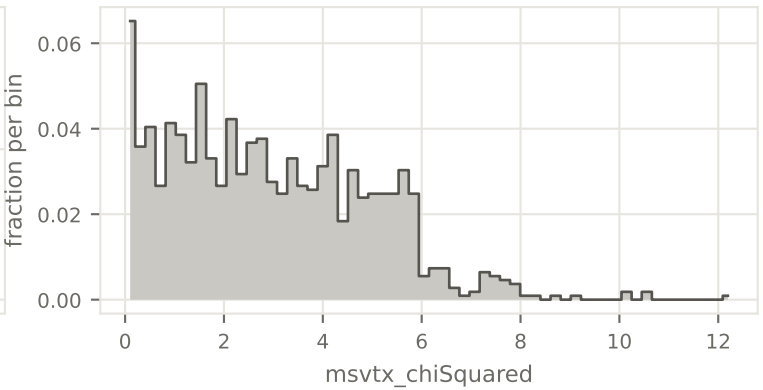


2 · combined signal
background n 1,089
combined, pooled D 0.220 n 4,987
equal weight per mass D 0.167

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

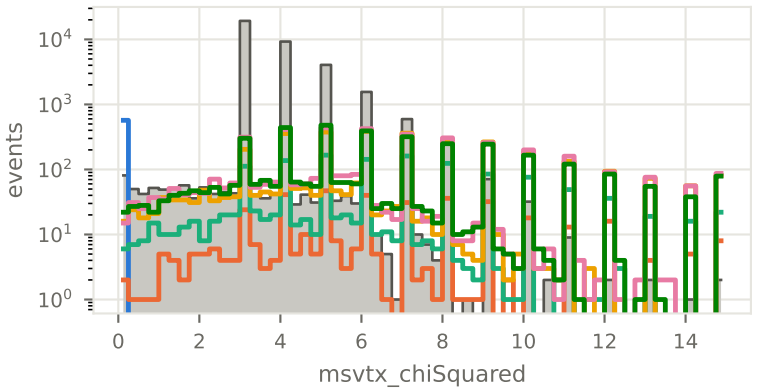
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_chiSquared

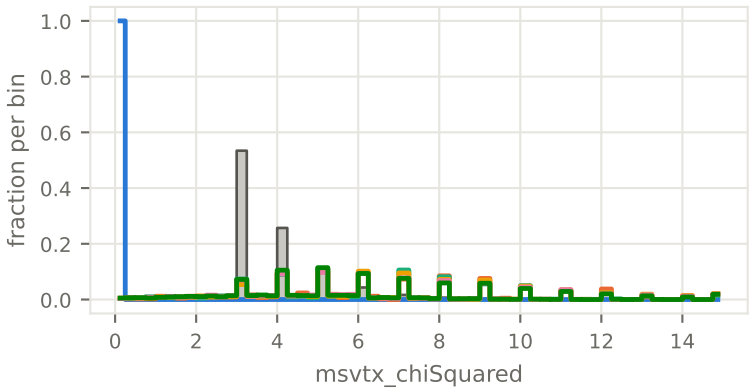
region: ALL · rank 17 / 98

D_equal 0.405 (noise 0.018) D_pooled 0.490 min 0.494 (mH1000_mS275) max 1.000 (mH125_mS15) bkg n = 36,059 sig n = 14,983

1a · each mass point vs background · raw

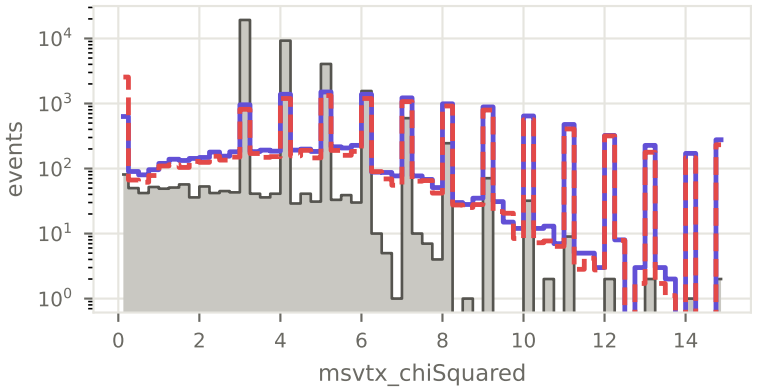


1b · each mass point vs background · normalized

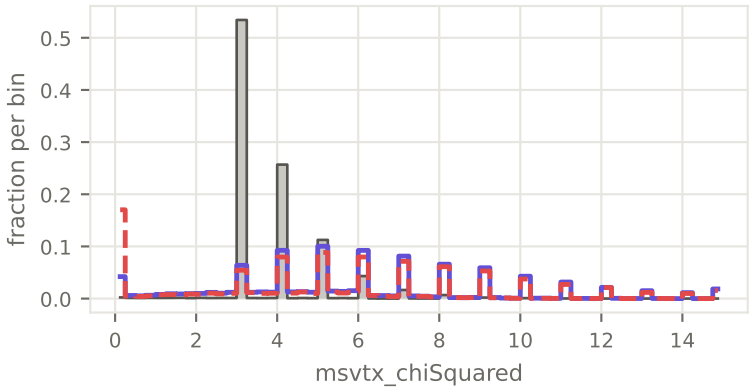


1 · each mass point
background n 36,059
mH125_mS15 D 1.000 n 570
mH125_mS55 D 0.550 n 423
mH200_mS50 D 0.525 n 1,520
mH400_mS100 D 0.544 n 3,792
mH600_mS150 D 0.509 n 4,490
mH1000_mS275 D 0.494 n 4,188

2a · combined signal vs background · raw

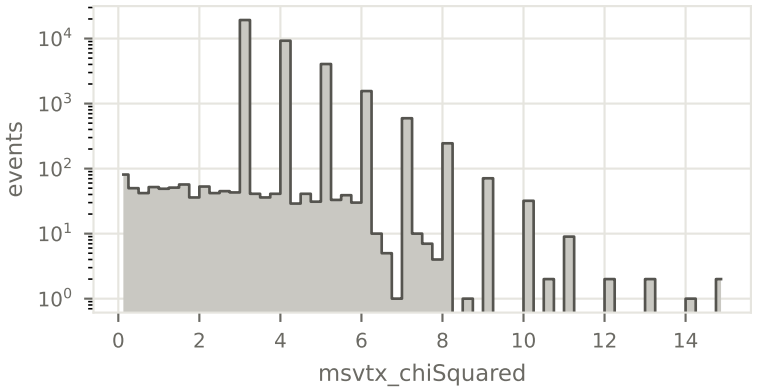


2b · combined signal vs background · normalized

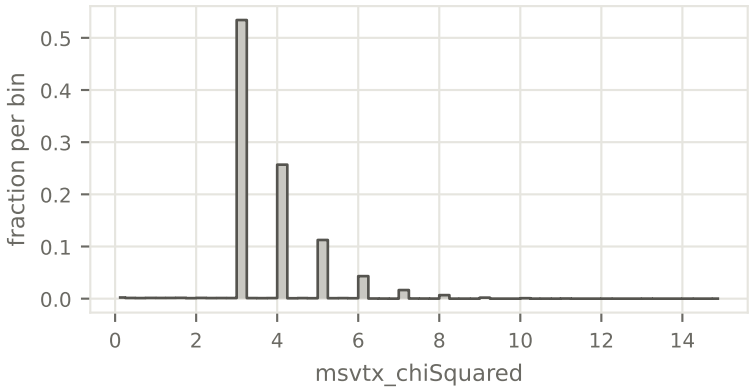


2 · combined signal
background n 36,059
combined, pooled D 0.490 n 14,983
equal weight per mass D 0.405

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 36,059

a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed